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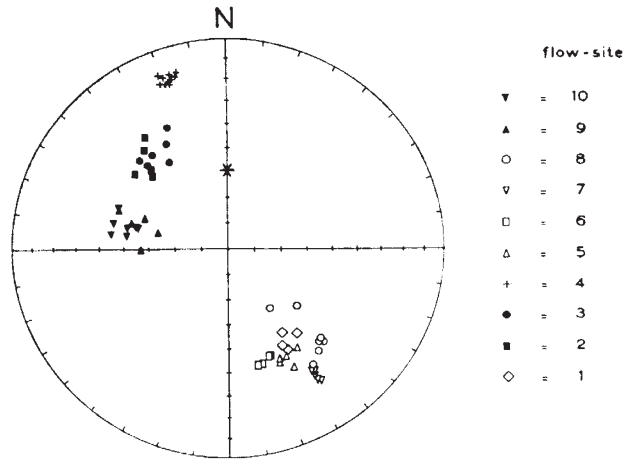


Fig. 3. Directions of magnetizations of the Alghero Volcanics after partial demagnetization in a.c. magnetic fields of 500 oersteds and with respect to the bedding planes. Open symbols denote north-seeking poles pointing up; full symbols denote north-seeking poles pointing down. ★ is direction of axial dipole field in the Alghero area.

Palaeomagnetism of the Alghero Trachyandesites

SARDINIA is situated about halfway between France and Italy in the Mediterranean. Geologists[1–3] have suggested that Sardinia has not always been in this position, but was once much nearer to France. From that original position, it is suggested that the island drifted to its present location either before[2] or after[1] the Middle Tertiary. The palaeomagnetic method is suited, given fulfilment of several conditions, to test such a hypothesis because drifting of Sardinia in the manner geologists have suggested could not have occurred without an anticlockwise rotation of the island. Rotation should be detectable by comparing palaeomagnetic directions from Sardinia with directions from continental Europe. In this communication, palaeomagnetic data of sixty hand samples from Middle Tertiary volcanic rocks near Alghero are used for comparison.

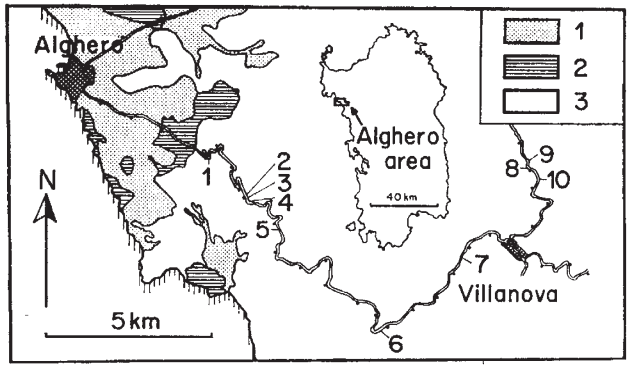


Fig. 1. Simplified geological map of the Alghero area (after Moretti *et al.*[6]) with sample sites. 1, Quaternary; 2, rock basement below Tertiary Volcanics (Triassic–Jurassic–Cretaceous rocks); 3, Tertiary Volcanics.

The sampled rocks are ignimbritic pheno-trachyandesites, which occur as extensive flow sheets. They are considered to be of Oligocene, perhaps Early Miocene age[4]. Tuffs and conglomerates are intercalated. Ten flows have been sampled from about twenty, each flow at one site only (Fig. 1). The higher the site number, the younger the flow: concerning flows 1–2, this is in accordance with Sanna[5]; concerning flows 2–10, with Moretti *et al.*[6].

The natural remanent magnetization (NRM) of the Alghero Volcanics was very similar in the sample groups of each flow. From each group, one pilot sample was progressively demagnetized[7,8] in alternating fields of up to 3,000 oersteds. The demagnetizations revealed that the total NRM of all these pilot samples was very similar and practically singular. Alternating fields below 150 oersteds removed a very small soft component (Fig. 2). All

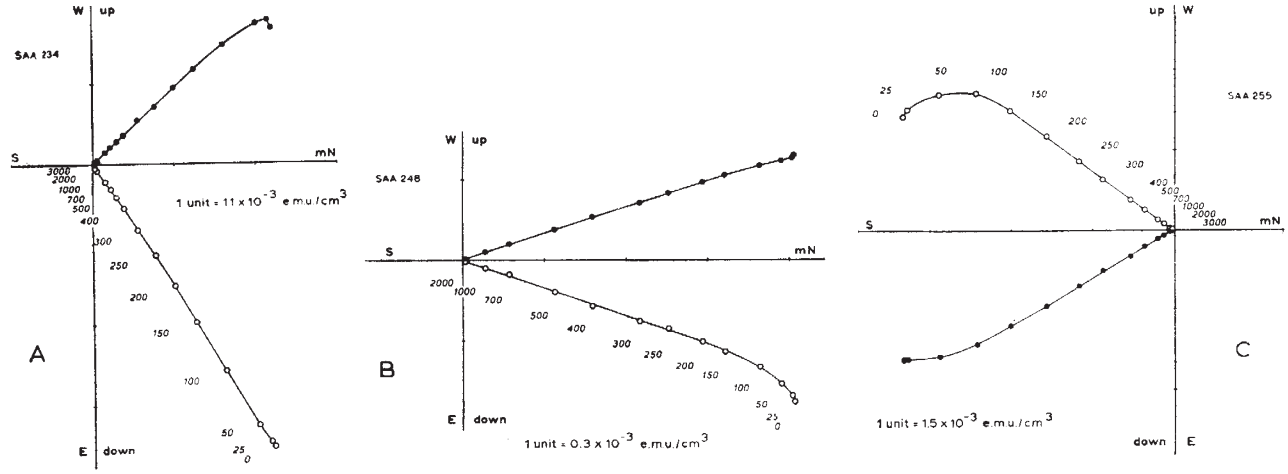


Fig. 2. Three examples of pilot sample demagnetization diagrams, respectively from flow 2 with normal magnetization (*A*), from flow 4 with normal magnetization and low inclination (*B*), and from flow 5 with reversed magnetization (*C*). —●— represent orthogonal projections of the successive positions of the terminus of the resultant magnetization vector during progressive demagnetization by a.c. magnetic fields. The numbers denote the intensities of these fields in oersteds. —○— indicate projections on the north-south vertical plane; full symbols indicate projections on the horizontal plane.

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Table 1. SUMMARY OF ALGHERO PALAEOMAGNETIC DATA

Flow-site	Strike and dip of bedding	Samples used/ collected	J $cm^{-3} \times 10^3$	Q	Direction of NRM Decl.	Incl.	Confidence α_{95}	Scatter K
1	327– 4	4/5	1·4– 1·8	9–14	148°	−47°	4·5°	386
2	327– 4	5/5	4·5– 5·2	7– 8	317°	+43°	6·5°	144
3	327– 4	6/6	2·2– 5·9	3–10	324·5°	+42·5°	6°	137
4	327– 4	11/11	0·2– 1·6	2– 7	341°	+16°	1·5°	1,090
5	7– 8	5/5	2·8– 7·4	6– 9	153°	−40°	4°	415
6	302–10	4/5	4·8–16·9	31–52	162·5°	−43°	3°	990
7	17– 7	5/5	0·5– 1·0	2– 4	146°	−28°	2·5°	1,090
8	0– 0	7/7	0·8– 1·3	2– 3	138·5°	−43°	9°	48
9	272–25	5/5	1·5– 1·9	3– 5	284°	+54·5°	8°	97
10	105–11	7/7	0·2– 0·6	1– 2	283°	+49°	4°	219

NRM directions after partial demagnetization with 500 oersted alternating field and with respect to the bedding planes.

remaining samples were subjected to alternating fields of 250, 500 and 1,000 oersteds and measured after each stage. The site mean directions were computed for the 500 oersted stage, which seemed to give the best representation of the original NRM. The palaeomagnetic results from the Alghero Volcanics are summarized in Fig. 3 and Table 1. The sampled rock section contains at least two normal and two reversed zones, and the mean virtual pole position is 54° N, 95° W ($\alpha_{95} = 12°$).

The characteristic NRM directions of the Alghero Volcanics have a systematic north-west–south-east declination (Fig. 3). Consequently the virtual poles are situated, in opposition to most other virtual poles of Middle Tertiary rocks from continental Europe, on the western hemisphere (Fig. 4). The three reversals found in the Alghero section indicate a relatively long period of flow deposition. It is therefore less plausible to attribute the observed deviation to ordinary secular variation. Hence these palaeomagnetic data from the Alghero Volcanics could be taken as the proof for a rotation (and drifting) of the Sardinian block after the Middle Tertiary. We do not consider the data to be definitely conclusive, because another explanation for the Alghero deviation might be polar wandering during the Middle Tertiary. Similar deviations on the European Continent have been reported by Roche[9] from Late Miocene lava flows from Auvergne (France). The age of the Alghero Volcanics, however, is considered to be Oligocene, perhaps Early Miocene[4], and

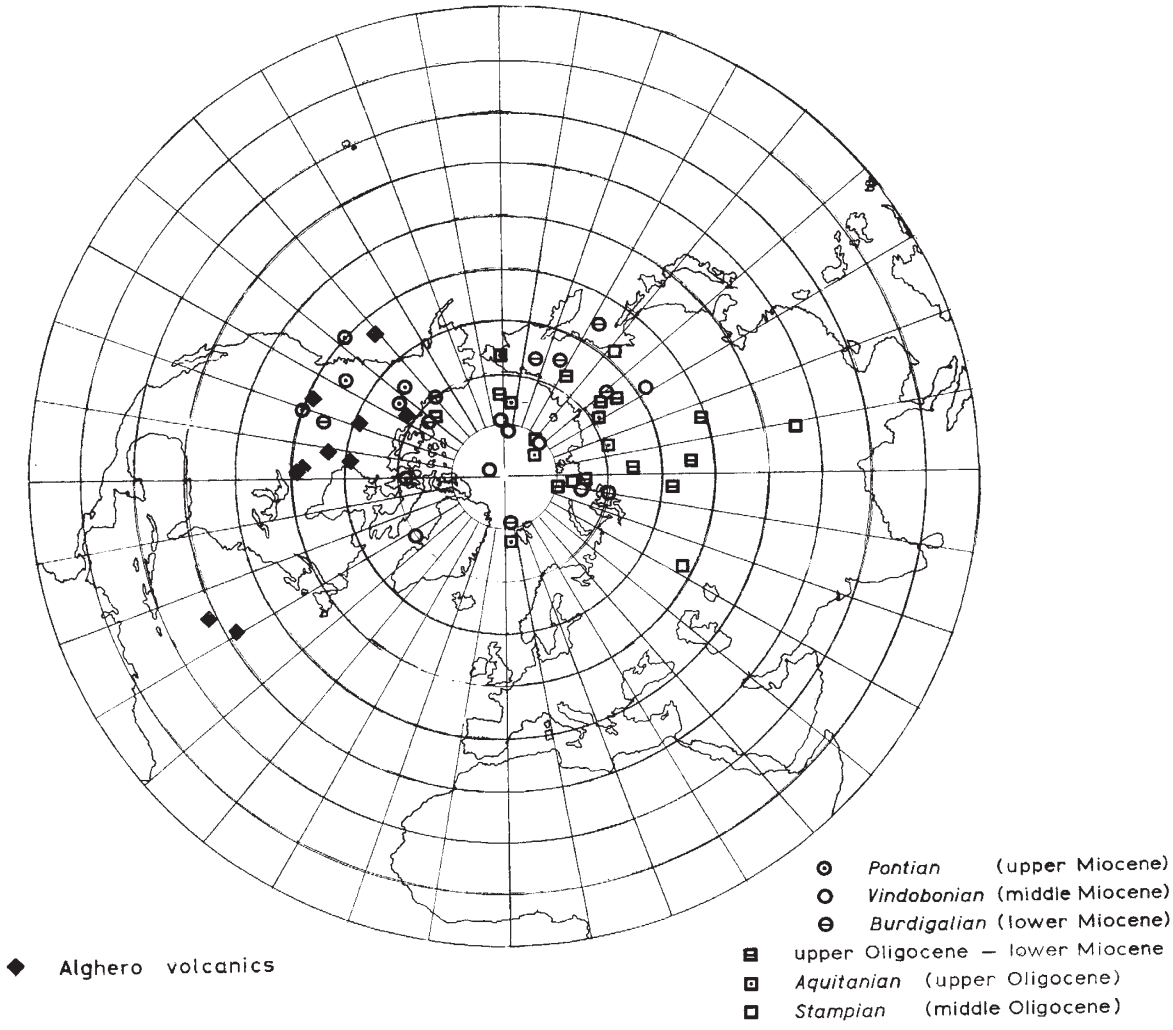


Fig. 4. Palaeomagnetic virtual pole positions of the lava flows sampled from the Alghero section in comparison with palaeomagnetic virtual poles calculated from the remanent magnetism reported for Oligocene and Miocene lavas from Western and Central Europe (Massif Central Auvergne[9,14–18], Kaiserstuhl[10,19], Siebengebirge and Westerwald[11,13], Northern Bohemia and Central Slovakia[12]). The stratigraphic names in the legend are in accordance with the nomenclature of the references.

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palaeomagnetic data of Middle Tertiary rocks from the European Continent have not yet produced indications for polar wandering during this period.

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BIOLOGICAL SCIENCES

Conformation of Human Serum High Density Lipoprotein and its Peptide Components

Circular dichroic (CD)[1] and optical rotatory dispersion (ORD)[2] measurements indicate that the protein moiety of human serum high density lipoproteins (HDL) contains a relatively high content of α-helical conformation. The low density lipoprotein fraction (LDL), on the other hand, has been found by CD and infrared spectroscopy to be relatively less helical in conformation and to contain considerable pleated sheet or β-structure[3–6]. The CD spectrum of HDL exhibits negative troughs at 222 and 208 nm, characteristic of α-helical conformation, while that of LDL has a major trough at 215–218 nm, suggestive of β-structure, and a shoulder at 208 to 210 nm compatible with some helical and/or random structure[8] (Table 1).

The infrared spectrum of LDL, between 1,600 and 1,700 cm^{-1} in $2H_2O$, has a major band at 1,617–1,625 cm^{-1} and shoulders at about 1,650 and 1,680–1,685 cm^{-1} (Table 1). Absorption bands in these regions have been associated with β, helical or disordered, and antiparallel chain β-conformation, respectively[8]. We now document the infrared spectrum of HDL in $2H_2O$ (Table 1), which contrasts with that for LDL in that the major band is at 1,637–1,640 cm^{-1} with shoulders at about 1,630 and 1,650 cm^{-1}. Removal of lipid from HDL did not significantly influence this pattern. If previous ORD[1] and CD[2] studies of HDL have correctly assessed its helical content, it would seem that the helical conformation of HDL-protein is manifested in an amide I infrared band in $2H_2O$ at a lower wavenumber than such conformation in proteins like myoglobin[8,9], and is similar in this regard to the helical polypeptide polyglycine II (ref. 10).

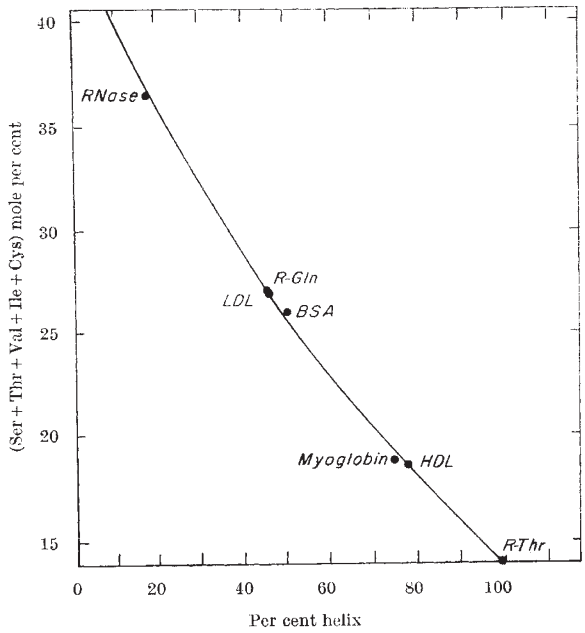


Fig. 1. Expected maximal helical contents of LDL, HDL, apoHDL-Gln (R-Gln) and apoHDL-Thr (R-Thr) based on their content of non-helical forming amino-acids. The points for these lipoprotein fractions represent only their content of the indicated amino-acids, obtained by total amino-acid analysis[12], and are not based on an optical estimation of helical content. The curve is replotted from the data of D. R. Davies[11], by courtesy of the author and the editor of the *Journal of Molecular Biology*.

Table 1. CD and infrared spectral characteristics of human serum low density (LDL) and high density (HDL) lipoprotein fractions*†

Fraction	CD spectrum (nm) Major trough	Shoulder	Infrared spectrum (cm^{-1}) Major band	Shoulders
LDL	215–218	208–210	1,617–1,625	1,640, 1,650 1,680–1,685
HDL	208, 220–222		1,637–1,640	1,630, 1,650

* Serum lipoprotein fractions were ultracentrifugally isolated[7,13] between densities 1·019–1·063 g/cm^3 (LDL), 1·063–1·21 g/cm^3 (HDL), and 1·125–1·195 g/cm^3 (HDL_3), and were immunochemically pure.
† CD and infrared spectra were measured as previously described[3]. All infrared spectra were measured in $2H_2O$.

For many proteins there is a negative correlation between the maximal helical content and the content of certain non-helical forming amino-acids[11]. On this basis one can predict that HDL would be approximately 70 to 80 per cent helical (Fig. 1), in reasonable agreement with the value calculated from ORD[2]. Consistent with the optical measurements just described, the maximal helical content of LDL would be expected to be significantly less—only about one-half that of HDL.

HDL contains several different polypeptides[13]. One of us (B. S.) has observed that two major HDL peptides constitute 90 per cent or more of the protein of HDL_3

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